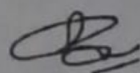
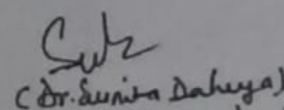


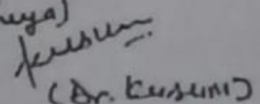
DEENBANDHU CHHOTU RAM UNIVERSITY OF SCIENCE & TECHNOLOGY
MURTHAL (SONEPAT) HARYANA-131039
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
B. Tech 2nd year Course
DIGITAL SYSTEM DESIGN
(Effective from Session 2025-26)

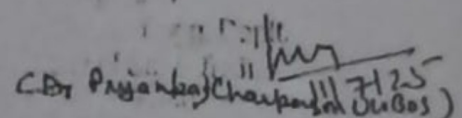
Course Code: ESECE201D	Internal Marks: Theory: 25; Lab: 15
Title of the Course: DIGITAL SYSTEM DESIGN	External Marks: Theory: 50; Lab: 10
L-T-P: 3-0-2	Maximum Marks: 100
Credits: 4	Duration of Exam: 03 Hours
About the Course: Digital system design is a foundational discipline in electronics and computer engineering, focusing on the creation of circuits that process binary information. These circuits underpin the functionality of all modern digital devices and enable the design of complex systems such as computers, digital watches, calculators, and embedded devices. This course will provide a comprehensive understanding of hardware description languages, digital system design principles.	
Course Objectives (CO): 1. To understand the fundamentals of Logic simplification. 2. To understand the concepts of MSI Devices and Applications. 3. To understand the concepts of Combinational and Sequential Logic Design, 4. To gain understanding of hardware description languages.	
Unit 1 (10 Lectures) Logic Simplification: Review of Boolean Algebra, Realization Using Gates. Karnaugh maps up to 6 variables, Quine-McCluskey method, VEM technique, Representation of Signed Number System, Binary Arithmetic, Code Conversion. Combinational Logic Design: Comparators, Multiplexers/Demultiplexers, Encoder, Decoder, Parallel Adders, Carry Look Ahead Adder, BCD Arithmetic, BCD Adder.	
Unit 2 (12 Lectures) Sequential Logic Design: Building blocks like S-R, J-K, D & T FF, Flip-Flop conversion, Shift registers: SISO, SIPO, PISO, PIPO, Bidirectional, Universal shift registers, Sequence Generator, Asynchronous and Synchronous counters: Operation; Up/Down counter, Mod-N counters, Ring counter, Johnson counter, BCD counter.	
Unit 3 (12 Lectures) Finite state machines: Introduction, Mealy & Moore types, Basic design steps, Design of synchronous FSM: Serial Binary Adder, Sequence detector, Parity Bit Generator, pulse train generator. Logic Families: Characteristics, Classification, TTL NAND gate, CMOS families and their interfacing, Basic working of TTL NAND gate and CMOS logic gate,	
Unit 4 (12 Lectures) Hardware Description Language: Overview of VHDL language, Data types and objects, different modeling styles in VHDL: Dataflow, Behavioral and Structural Modeling, Implementation of combinational and sequential circuits using VHDL.	



(Dr. Lakshmi Yadav)


(Dr. Sunil Dahiya)


(Dr. Kusum)


(Dr. Pranjana Choudhary)